

MADHAV KUMAR YADAV

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🐙 GitHub

🌐 LinkedIn

📁 Portfolio

Summary

Analytical and tech-savvy Data Scientist and Machine Learning Engineer with strong communication skills and expertise in statistical modeling, machine learning, data analytics, and MLOps. Builds end-to-end ML pipelines: data preprocessing, feature engineering, model training, and deployment using Docker, FastAPI, and Streamlit. Proficient in Python, Scikit-learn, and cloud deployment. Strong foundation in algorithms, applied mathematics, linear algebra, and calculus. Demonstrates leadership by guiding project direction and collaborating effectively.

Skills

Technical: Python, SQL, MySQL, Java (Basic), HTML/CSS, Scikit-learn, Pandas, NumPy, EDA, Hyperparameter Tuning, Feature Engineering, Git, Docker, FastAPI, Flask, Streamlit, MongoDB, MLflow, Power BI, Tableau, MLOps, REST APIs, Statistical Analysis, Statistics, Machine Learning, Data Analytics, Algorithms, Applied Mathematics, Data Visualization, Business Rules, Linear Algebra, Researching, Calculus.

Soft: Analytical, Communication Skills, Tech Savvy, Problem-Solving, Teamwork, Leadership, Drive and Initiative.

Projects

Network Security – Phishing & Intrusion Detection

Jan 2026

🐙 GitHub Repo

- Led the development of ML classification models achieving **96% accuracy** in detecting phishing websites and malicious network activity through feature engineering and hyperparameter tuning.
- Integrated MLflow for experiment tracking (50+ runs) and MongoDB for data storage, improving workflow reproducibility by 40%.
- Containerized the application with Docker and deployed on Streamlit, enabling batch prediction of 10,000+ records in under 2 seconds.
- *Tech: Python, Scikit-learn, Streamlit, MLflow, MongoDB, Docker*

Forest Fire Weather Index (FWI) Prediction

Oct 2025

🐙 GitHub Repo

- Developed Flask web app predicting Forest Fire Weather Index with **92% accuracy**, reducing fire risk assessment time by 60%.
- Performed EDA on Algerian Forest Fire dataset (244 instances, 11 features) to identify key fire triggers, then tuned Linear Regression using GridSearchCV (5-fold CV).
- Deployed on Render, handling 500+ real-time predictions per day with 99.9% uptime.
- *Tech: Python, Scikit-learn, Flask, Pandas, Render*

Student Performance Predictor (MLOps Pipeline)

Dec 2025

🐙 GitHub Repo

- Built end-to-end MLOps pipeline predicting student outcomes with **88% F1-score**, processing 5,000+ student records.
- Designed automated data preprocessing (handling 15+ features) and model training modules; containerized with Docker and exposed REST APIs via FastAPI (latency <100ms).
- Implemented logging and monitoring (MLflow + custom logger), reducing inference latency by 30% (from 150ms to 105ms).
- *Tech: Python, Scikit-learn, FastAPI, Docker, Pandas*

Education

Bachelor of Technology in Computer Science

2023 – 2027

Jagannath University Jaipur, Chaksu, India

CGPA: 7.7/10

Relevant Coursework: Machine Learning, DSA, DBMS, Probability & Statistics, Big Data Analytics, Linear Algebra, Calculus, Applied Mathematics.

Certifications

- Microsoft Power BI (Udemy)
- Introduction to Cybersecurity (Cisco Networking Academy)
- Python Essentials 1 (Cisco Networking Academy)
- HackerRank 5 Stars (C++ & SQL)